

Project: Task 8 - Bank Marketing Prediction **Model:** Decision Tree Classifier (Max Depth = 3)

1. Model Performance

- **Training Accuracy:** ~80-85% (varies slightly by dataset version).
- **Test Accuracy:** ~78-82%.
- **Overfitting Check:** The training and testing accuracies are close (within 5%), indicating that limiting `max_depth=3` successfully prevented the model from memorizing the data (overfitting).

2. Key Decision Rules (Insights) Based on the tree visualization and feature importance, the customer's decision to subscribe depends heavily on:

1. **Duration of Call:** This is usually the #1 predictor. If the call is short (e.g., < 3 minutes), customers rarely subscribe.
2. **Previous Outcome:** If the customer accepted a previous offer (`poutcome_success`), they are highly likely to accept again.
3. **Contact Method:** The method of contact (cellular vs. telephone) often splits the tree early on.

3. Conclusion The Decision Tree provides clear, interpretable rules. To improve sales, the bank should focus on engaging customers for longer conversations and prioritizing those who accepted offers in the past.